

Data Science

CSE558

Recap

- Linear Algebra Basics
- Dimensionality Reduction

The diagram illustrates the Singular Value Decomposition (SVD) of a matrix A . It shows a large blue square representing matrix $A_{m \times n}$ on the left. To its right is an approximation symbol $\approx$. Further right is a tall orange rectangle representing matrix $U_{m \times r}$. To the right of U is a small purple square representing the diagonal matrix $\Sigma_{r \times r}$. Finally, to the right of Σ is a wide green rectangle representing the matrix $V^T_{r \times n}$. The entire equation is represented as $A_{m \times n} \approx U_{m \times r} \Sigma_{r \times r} V^T_{r \times n}$.

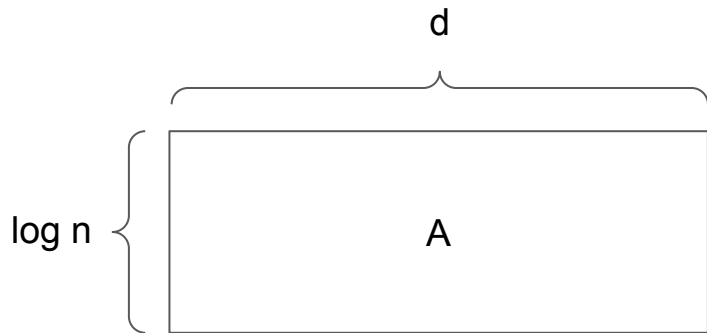
Overview

- Johnson Lindenstrauss

Johnson Lindenstrauss (1984)

For any set of n points in d -dimension. There is a family of matrices $\mathcal{A}$ such that for every $A \in \mathcal{A}$, pair of points (x, y) and $\varepsilon \in (0, 1)$ with a very high probability

$$(1 - \varepsilon)\|x - y\|_2^2 \leq \|Ax\|_2^2 - \|Ay\|_2^2 \leq (1 + \varepsilon)\|x - y\|_2^2$$



Dimension Reduction

Random projection

- Oblivious to the data
- Preserves lengths of points
- Preserves pairwise distance between points

Input: n points in d -dimension.

Time to compute all pair distance: n^2d

Sampling 'A' from ' $\mathcal{A}$ '

Independent gaussian entries

A

Why oblivious to data?

Data oblivious algorithms in past?

Input: n points in d -dimension.

Time to compute all pair distance: $O\left(\frac{nd \log(n)}{\varepsilon^2} + \frac{n^2 \log(n)}{\varepsilon^2}\right)$

JL reduces to Norm Preservation

Let A be a random matrix. For any vector v if,

$$P \left((1 - \varepsilon) \|v\|_2^2 \leq \|Av\|_2^2 \leq (1 + \varepsilon) \|v\|_2^2 \right) \geq 1 - \frac{1}{n^3}$$

then A also guarantees JL theorem for any set of n vectors.

Fact: If $n_1 \sim \mathcal{N}(\mu_1, \sigma_1^2)$ and $n_2 \sim \mathcal{N}(\mu_2, \sigma_2^2)$ are two independent gaussian random variables, then

$$n_1 + n_2 \sim \mathcal{N}(\mu_1 + \mu_2, \sigma_1^2 + \sigma_2^2)$$

Proof: JL Lemma (Unbiased Estimate)

Let A be a $k \times d$ matrix, whose entries are iid samples from $N(0, 1/k)$.

For a fix vector d -dimensional vector ' x ' analyze $\|Ax\|^2$

$$\left\| \begin{bmatrix} \text{---} \\ \text{---} \\ \text{---} \\ \text{---} \\ \text{---} \\ \text{---} \end{bmatrix} \times \begin{bmatrix} \text{---} \\ \text{---} \\ \text{---} \\ \text{---} \\ \text{---} \\ \text{---} \end{bmatrix} \right\|^2 = \sum \left[\underbrace{\begin{bmatrix} \text{---} \\ \text{---} \\ \text{---} \\ \text{---} \\ \text{---} \\ \text{---} \end{bmatrix} \times \begin{bmatrix} \text{---} \\ \text{---} \\ \text{---} \\ \text{---} \\ \text{---} \\ \text{---} \end{bmatrix}}_{r \sim \mathcal{N}\left(0, \frac{\|x\|_2^2}{k}\right)} \right]^2$$

The diagram illustrates the proof of the JL Lemma. It shows the squared norm of the product of a $k \times d$ matrix A and a d -dimensional vector x . The matrix A is represented by a rectangle with 6 horizontal lines, and the vector x is represented by a vertical rectangle. The product Ax is shown as a vertical rectangle with 6 horizontal lines, labeled $N(0, 1/k)$. The squared norm $\|Ax\|^2$ is then expressed as the sum of the squared norms of the rows of Ax . Each row of Ax is a scalar value, and the sum of their squares is the squared norm. The diagram uses a large bracket to group the rows of Ax , and a brace below it to indicate that the distribution of the rows is $r \sim \mathcal{N}\left(0, \frac{\|x\|_2^2}{k}\right)$.

Proof: JL Lemma (Unbiased Estimate)

Let A be a $k \times d$ matrix, whose entries are iid samples from $N(0, 1/k)$.

For a fix vector d -dimensional vector ' x ' analyze $\|Ax\|^2$

$$\left[\underbrace{\left[\text{N}(0, 1/k) \right] \times \left[\begin{array}{c} \vdots \\ x \end{array} \right]}_{r_1 \sim \mathcal{N}\left(0, \frac{\|x\|_2^2}{k}\right)} \right]^2 + \dots + \left[\underbrace{\left[\text{N}(0, 1/k) \right] \times \left[\begin{array}{c} \vdots \\ x \end{array} \right]}_{r_k \sim \mathcal{N}\left(0, \frac{\|x\|_2^2}{k}\right)} \right]^2$$

$$\mathbf{E}[\|Ax\|_2^2] = \mathbf{E}\left[\sum_{i \leq k} r_i^2\right] = \|x\|_2^2$$

Proof: JL Lemma (Bound Failure Probability)

Let $y_i = \frac{r_i}{\|x\|_2/\sqrt{k}}$

Notice $y_i \sim \mathcal{N}(0, 1)$

$$\mathbf{P} \left(\sum_{i \leq k} r_i^2 \geq (1 + \varepsilon) \|x\|_2^2 \right) = \mathbf{P} \left(\sum_{i \leq k} y_i^2 \geq (1 + \varepsilon) k \right) \leq e^{\frac{-k\varepsilon^2}{8}}$$

Setting, $k = \frac{24 \log(n)}{\varepsilon^2}$

$$\mathbf{P} \left(\sum_{i \leq k} r_i^2 \geq (1 + \varepsilon) \|x\|_2^2 \right) \leq \frac{1}{n^3}$$

Extension

$$\text{entries} \sim \left\{ \frac{-1}{\sqrt{k}}, \frac{+1}{\sqrt{k}} \right\} \text{ with } \\ \text{equal probability}$$

A

$$\text{entries} \sim \begin{cases} \sqrt{\frac{3}{k}} \cdot (-1), \text{ w.p } 1/6 \\ 0, \text{ w.p } 2/3 \\ \sqrt{\frac{3}{k}} \cdot (+1), \text{ w.p } 1/6 \end{cases}$$

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$$\mathbf{P} \cdot \mathbf{H} \cdot \mathbf{D}$$

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Sharp contrast to PCA

Applications

k-means clustering

Linear regression